

T. Dilli praveeth Podd  
192492377

## SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

7A  
100  
18

### COURSE OUTCOME COVERED

**CO1:** Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

#### Description about the assessment tool:

Short-answer questions effectively measure conceptual understanding, logical reasoning, and the ability to apply theoretical knowledge without requiring lengthy descriptive responses. This assessment method is also efficient for evaluating multiple OSI-related concepts within a limited time while ensuring objective and consistent evaluation.

### Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.
2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every

window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

## RUBRICS FOR CALCULATION

| Criteria                  | Weightage | Level 4 – Exemplary                                         | Level 3 – Proficient                    | Level 2 – Developing                         | Level 1 – Beginning                      |
|---------------------------|-----------|-------------------------------------------------------------|-----------------------------------------|----------------------------------------------|------------------------------------------|
| Understanding of Concepts | 25%       | Demonstrates a thorough, accurate understanding of concepts | Good understanding with minor gaps      | Basic understanding with some misconceptions | Poor understanding; major misconceptions |
| Explanation               | 25%       | Detailed, well-explained answers with relevant examples     | Clear explanation with adequate details | Limited explanation; lacks depth             | Poor or unclear explanation              |
| Application               | 20%       | Effectively applies concepts with strong analytical ability | Applies concepts with minor errors      | Limited application with weak analysis       | No application or incorrect analysis     |
| Organization              | 15%       | Well-structured answers with logical flow and coherence     | Organized with minor issues in flow     | Basic structure, lacks clarity               | Poor organization, incoherent answers    |
| Accuracy                  | 10%       | Completely accurate and covers all required points          | Mostly accurate with minor omissions    | Partially correct with missing points        | Incorrect answers with major gaps        |
| Clarity                   | 5%        | Clear, neat, professional presentation                      | Understandable with minor issues        | Limited clarity, readability issues          | Poorly written, difficult to understand  |

# 1. File Transfer through Transport & Data Link layers.

Given

\* File size = 150 MB =  $150 \times 10^6 = 150,000,000$  bytes

\* Bandwidth = 50 Mbps

\* Segment size = 1500 bytes

\* Data link overhead = 40 bytes/frame

Calculations

Number of Segments

$$\frac{150,000,000}{1500} = 100,000 \text{ Segments.}$$

Each segment becomes one frame.

Number of frames = 100,000 frames.

Total Data Link overhead

$$100,000 \times 40 = 4,000,000 \text{ bytes} = 4 \text{ MB}$$

Total Data sent

$$150 + 4 = 154 \text{ MB}$$

Transmission Time

$$154 \text{ MB} = 1232 \text{ Mb}$$

$$\frac{1232}{50} = 24.64 \text{ Seconds.}$$

Answer

- Segments = 100,000
- Frames = 100,000
- Data link overhead = 4 MB

- Total Transmission Time  $\approx 24.64s$

### Reliability:-

- Transport Layer: User segmentation, acknowledgment, sequencing, retransmissions and error recovery.
- Data Link Layer: Detects frame errors using CRC, control flow, and retransmits corrupted frames.

## 2. Sliding window protocol:-

Given

- Window Size = 6 frames
- Total Frames = 48
- Frame Size = 1024 bytes

### Calculations:-

Transmission windows

$$\frac{48}{6} = 8.$$

### Retransmissions.

One frame lost in each window

$$8 \times 1 = 8$$

### Answer

- windows Required = 8
- Retransmissions = 8 frames.

## IP Fragmentation

Given

- Packet Size = 1500 bytes
- MTU = 1500 bytes
- IP header = 20 bytes

### Calculations

Maximum payload per fragment

$$1500 - 20 = 1480 \text{ bytes.}$$

Number of fragments

$$\lceil 1500 / 1480 \rceil = 2$$

Header overhead

$$2 \times 20 = 40 \text{ bytes.}$$

### Answer:-

- Fragments created = 2
- Header overhead = 40 bytes.

### Network Layer:-

The Network Layer fragments oversized packets into each fragment independently, and the destination reassembles them using fragment offsets and identification Number.

#### 4. Sliding window protocol (Repeated Question)

Answer

- Windows = 8
- Retransmissions = 2 times

Flow control

Flow control synchronizes sender and receiver speeds, improving bandwidth utilization and minimizing congestion.

#### 5. Session Layer Synchronization

Given

- File Size = 600 MB
- Checkpoint Every = 60 MB.
- Failure After = 390 MB.

Calculations:-

Checkpoints before failure

$$390 / 60 = 6.5$$

Completed checkpoints = 6

Last checkpoint = 360 MB

Retransmission after recovery

$$390 - 360 = 30 \text{ MB}$$

- Checkpoint created = 6
- Data Retransmitted = 30 MB

## Presentation Layer compression & Encryption

Given

- Original File = 900 MB
- Compression = 40%
- Encryption overhead = 5%

### Calculation

Compressed size

$$900 \times 0.60 = 540 \text{ MB}$$

Encrypted size

$$540 \times 1.05 = 567 \text{ MB}$$

Overall Reduction

$$\frac{900 - 567}{900} \times 100 = 37\%$$

### Answer

- Compressed size = 540 MB
- Encrypted size = 567 MB
- Overall Reduction = 37%

### Presentation Layer:-

This layer compress data to reduce transmission time and encrypts it to maintain confidentiality and security.

## \* FTP Data Transfer

Given

- File size = 3GB = 3000 MB
- Request size = 3MB
- Throughput = 120 Mbps.

Calculations:-

Requests

$$3000/3 = 1000$$

Transmission Time

$$3000 \text{ MB} = 24000 \text{ Mb}$$

$$24000/120 = 200 \text{ seconds.}$$

Answer

- Requests = 1000
- Transmission Time = 200 Seconds.

Application Layer

Provides user-friendly services like FTP, file management authentication, and reliable data transfer.

## 8. End-to-End Delay:-

Given:-

Processing Delay.

- Application = 4ms
- Transport = 3ms
- Network = 5ms
- Data Link = 2ms

physical = 1ms

propagation Delay = 18ms

Transmission Delay = 12ms

calculations:-

processing Delay

$$4 + 3 + 5 + 2 + 1 = 15 \text{ ms}$$

Total Delay

$$15 + 18 + 12 = 45 \text{ ms}$$

Answer

- Total End-to-End Delay = 45ms.

Layer contributions:-

- Application : User services
- Transport : Reliable delivery.
- Network : Routing
- Data Link : Frame transmission.
- physical : signal transmission.

9. Protocol overhead

Given

- used data = 80 MB = 80,000,000 bytes
- frame size = 1600 bytes
- overhead = 80 bytes.

Calculations:-

use for payload

$$1600 - 80 = 1520 \text{ bytes.}$$

Frames

$$80000000 / 1520 \approx 52632$$

Total overhead

$$52632 \times 80 = 4210560 \text{ bytes} \approx 4.21 \text{ MB}$$

Efficiency

$$\frac{1520}{1600} \times 100 = 95\%$$

Answer

- Frames = 52632
- Overhead = 4.21 MB
- Transmission Efficiency = 95%

10. Healthcare IoT Data Transfer

Given

- Original Data = 240 MB
- Compression = 30%
- Segment Size = 1400 bytes
- Data Link overhead = 60 bytes/frame.

Calculations

Compressed size

$$240 \times 0.70 = 168 \text{ MB}$$

Compressed Data

$$168,000,000 \text{ bytes}$$

Segments

$$168000000 / 1400 = 120000$$

Protocol overhead

$$120000 \times 60 = 7200000 \text{ bytes} = 7.2 \text{ MB}$$

Final Data sent

$$168 + 7.2 = 175.2 \text{ MB}$$